🚘 Vehicle Accident Alert System with Multiple Emergency Calling

📌 Project Overview

The **Vehicle Accident Alert System with Multiple Emergency Calling** is an embedded safety system designed to automatically detect vehicle accidents and notify multiple emergency contacts.
The system uses an **accelerometer-based impact detection mechanism**, along with **GPS and GSM communication**, to ensure rapid emergency response by sending location-based alerts and initiating phone calls.

This system is especially useful for **two-wheelers and four-wheelers**, where immediate assistance after an accident can be life-saving.

🎯 Project Objectives

- Detect sudden vehicle impacts using an accelerometer
- Automatically fetch real-time GPS location
- Send SMS alerts containing Google Maps location
- Make emergency phone calls to **multiple contacts**
- Provide local audio alert using a buzzer
- Allow manual system reset to prevent false alerts

🧠 System Description

The system continuously monitors acceleration values from the ADXL345 sensor. When a sudden abnormal acceleration change is detected beyond a predefined threshold, the system assumes an accident has occurred and initiates the alert sequence.

After detection:

1. GPS location is captured
2. Emergency calls are placed sequentially

3. SMS alerts with location are sent to each contact

🧩 Hardware Components

Component	Description
Arduino UNO / Nano	Main microcontroller
ADXL345 Accelerometer	Accident detection
SIM800L GSM Module	Calling and SMS alerts
NEO-6M GPS Module	Location tracking
Buzzer	Local alert indication
Push Button	Manual system reset
Power Supply	5V (regulated for GSM)

🔌 Pin Configuration

GSM Module (SIM800L)

- RX → D3
- TX → D2

GPS Module (NEO-6M)

- RX → D9
- TX → D10

Other Connections

- Buzzer → D12
- Reset Button → D11

🧑‍💻 Software & Libraries Used

- Arduino IDE
- SoftwareSerial
- TinyGPS++
- Adafruit ADXL345 Library
- Adafruit Unified Sensor Library
- Wire Library
- Math Library

⚙️ Working Principle

1️⃣ Impact Detection

- ADXL345 measures acceleration along X, Y, and Z axes
- Change in acceleration magnitude is calculated
- If magnitude exceeds the sensitivity threshold → **Accident detected**

2️⃣ GPS Location Acquisition

- GPS module attempts satellite lock
- Latitude and longitude are extracted
- Location is formatted into a Google Maps link

3️⃣ Emergency Alert Sequence

Once an accident is detected:

- Buzzer turns ON for immediate alert
- System waits for a short delay
- Emergency Contact 1:
    - Phone call is made

    - SMS with location is sent
- Emergency Contact 2:
    - Phone call is made
    - SMS with location is sent

4️⃣ Manual Reset

- Push button resets the alert state
- Helps prevent false emergency notifications

📩 SMS Alert Format

Accident Alert! Location: http://maps.google.com/maps?q=loc :,

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🚨 Key Features

- Automatic accident detection
- Multiple emergency contact support
- Real-time GPS location sharing
- Emergency calling + SMS alert
- Manual reset mechanism
- Compact and low-cost design

📈 Advantages

- Faster emergency response
- Works without internet (GSM-based)
- Scalable for more contacts
- Suitable for real-world deployment
- Simple and reliable logic

⚠️ Limitations

- GPS signal may be weak indoors or in tunnels
- GSM network availability required
- Power regulation critical for SIM800L
- Fixed sensitivity may cause false triggers on rough roads

🔮 Future Enhancements

- Dynamic sensitivity adjustment
- IoT cloud integration
- Mobile application support
- Accident severity classification
- Vehicle speed analysis
- Battery backup system

🚘 Applications

- Personal vehicle safety systems
- Two-wheeler accident monitoring
- Fleet safety management
- Emergency response automation
- Smart transportation systems

👤 Author

Sahad Nisham K
Instagram: https://www.instagram.com/sahadnisham.k/
LinkedIn: https://www.linkedin.com/in/sahad-nisham-k/

▦ Conclusion

This project demonstrates a reliable and efficient embedded solution for accident detection and emergency alerting.
By combining sensor-based detection with GSM and GPS technologies, the system ensures timely assistance and enhances road safety.